Checkpoint 6 Science — Practice Test 3

Paper 1: 35 minutes · calculator allowed · Paper 2: 35 minutes · Total 80 marks

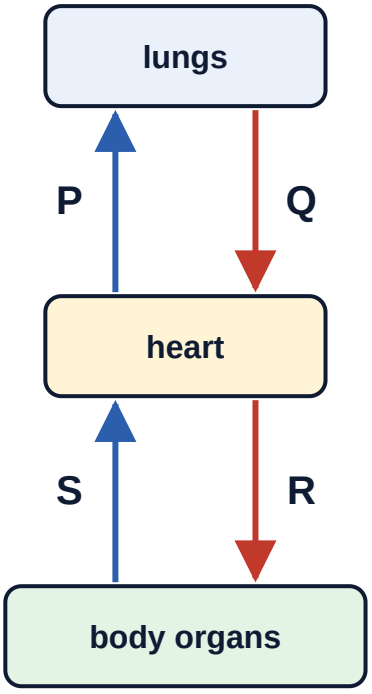
Answer all questions. Write your answers in the spaces provided. Show your working where there is space for it. The number of marks for each question is shown in brackets [ ].

Name _____ Class _____ Date _____ Mark: ____ / 80

Abdelkriem Academy · practice paper written in the style of Cambridge Checkpoint. Not an official Cambridge paper.

Paper 1 — 35 minutes — 40 marks

1 Hana visits the doctor's clinic for a check-up. The doctor shows her this diagram of how blood flows around the body. The arrows show which way the blood flows. **[1]**



(a) Which letter shows the blood vessel that carries blood from the heart to the body organs?

(b) What **type** of blood vessel is R? **[1]**

(c) Inside the body organs, blood flows through tiny blood vessels with walls only one cell thick. Name these blood vessels. **[1]**

(d) The nurse counts Hana's pulse for **30 seconds**. She counts 38 beats. What is Hana's pulse rate in beats per minute? **[1]**

Working

Answer ____ beats per minute

(e) Blood carries oxygen around the body. Name **two** other things that blood transports. **[2]**

2 The doctor sees three patients. **[1]**

Patient	Illness	What caused it
Malak	athlete's foot (itchy skin between the toes)	a fungus
Karim	throat infection with a fever	bacteria
Seif	malaria	a parasite carried by mosquitoes

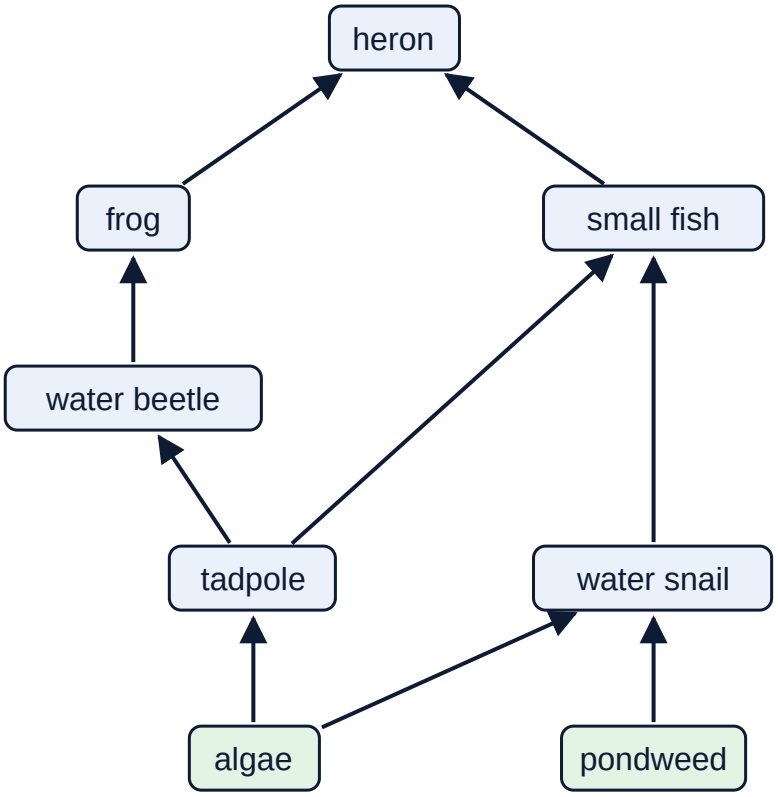
(a) The next patient has **flu**. What causes flu?

- ☐ A) a fungus
- ☐ B) a virus
- ☐ C) a parasite
- ☐ D) bacteria

(b) Karim breathed in the bacteria. Explain how **mucus** in his nose and airways helps to protect his lungs. **[1]**

(c) Seif got malaria from a mosquito bite. Give **two** ways a person can avoid being bitten by mosquitoes. **[2]**

3 A class studies a pond. They draw this food web. **[1]**



(a) How many food chains are there in this food web?

Working

Answer

(b) Name **one** producer in this food web. **[1]**

(c) What is the source of energy for this food web? **[1]**

(d) Write down the food chain from this food web that has **five** living things. **[1]**

(e) A poison from a nearby field washes into the pond. The algae take in the poison. It is not broken down in living things. Which animal will end up with the **most** poison in its body? **[1]**

- ☐ A) tadpole
- ☐ B) water snail
- ☐ C) water beetle
- ☐ D) heron

(f) Explain your answer to part (e). **[1]**

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4 In the school laboratory, Leila investigates how the temperature of water affects the time a piece of chocolate takes to melt. She puts a 10 g square of chocolate in a small metal dish and floats the dish in a water bath. She times how long the chocolate takes to melt completely. **[1]**

Temperature of water in °C	Time to melt in seconds
40	240
50	150
60	190
70	70
80	50

(a) What is the **independent** variable in Leila's investigation?

(b) Give **one** variable Leila must keep the same to make this a fair test. **[1]**

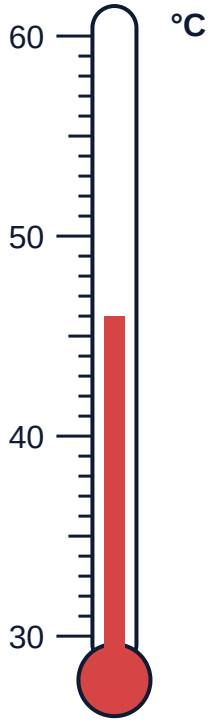
(c) One result does not fit the pattern (it is anomalous). At which temperature is the anomalous result? **[1]**

Working

Answer ☐ °C

(d) Describe the pattern in Leila's results. **[1]**

[1]



(e) Before one test, Leila reads the thermometer in the water bath. What temperature does it show?

Working

Answer °C

(f) Melting chocolate is a **reversible** change. Describe how Leila could show this. **[1]**

(g) Leila puts a piece of candle wax in water at 40 °C. The chocolate melted at this temperature, but the wax does not melt. Which statement is correct? **[1]**

- ☐ A) Wax has a lower melting point than chocolate.
- ☐ B) Wax and chocolate have the same melting point.
- ☐ C) Wax has a higher melting point than chocolate.
- ☐ D) Wax can never melt.

5 Ziad visits a science museum. A large model shows the Sun and the planets of the Solar System. **[1]**

(a) Which planet is the **fourth** planet from the Sun?

- ☐ A) Mars
- ☐ B) Venus
- ☐ C) Jupiter
- ☐ D) Earth

(b) Which sentence about movement in the Solar System is correct? **[1]**

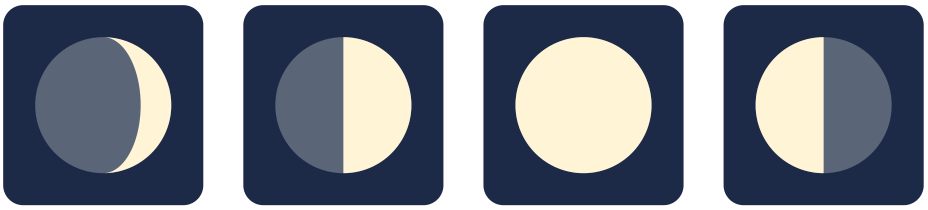
- ☐ A) The Sun orbits the Earth once a year.
- ☐ B) The Moon orbits the Sun once a month.
- ☐ C) The Earth orbits the Sun about once every 365 days.
- ☐ D) The planets orbit the Moon.

(c) What does the Moon orbit? **[1]**

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6 (continued) **[1]**

A poster in the museum shows the Moon as seen from Egypt on four different nights.



(d) On day 0 there is a new moon. Which drawing shows the Moon about **7 days** later?

- ☐ A) A
- ☐ B) B
- ☐ C) C
- ☐ D) D

7 Omar rides his bicycle along a flat road. The diagram shows the two horizontal forces on Omar and his bicycle. **[1]**



(a) Work out the overall (resultant) force on Omar and his bicycle.

Working

Answer N

(b) What happens to Omar's speed? **[1]**

- ☐ A) It stays the same.
- ☐ B) It increases.
- ☐ C) It decreases.
- ☐ D) He stops at once.

(c) Later, the forward force is 100 N and the backward forces add up to 100 N. Which describes Omar's movement now? **[1]**

- ☐ A) He slows down and stops.
- ☐ B) He speeds up.
- ☐ C) He moves backwards.
- ☐ D) He keeps moving at a steady speed.

(d) Omar squeezes his brakes. Rubber brake pads press on the wheels. Name the force between the brake pads and the wheels that slows the wheels down. **[1]**

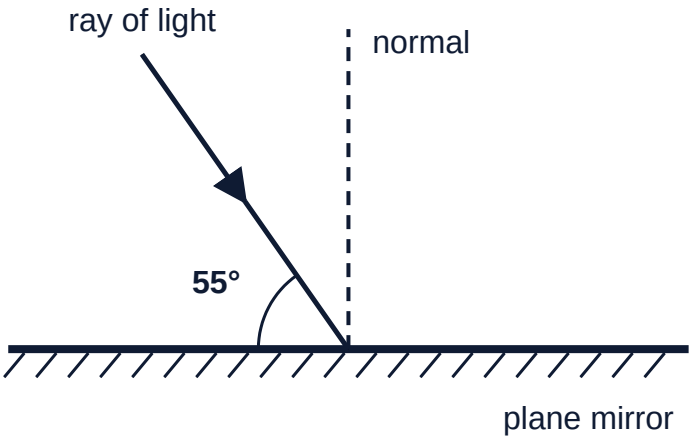
(e) Omar stops. He and his bicycle are now still on the flat road. The weight of Omar and his bicycle is 550 N. What is the size of the upward force from the road? **[1]**

Working

Answer N

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8 In the museum's light room, Ziad shines a ray of light at a plane mirror. The dashed line is the normal. **[1]**



(a) What is the angle of incidence?

Working

Answer °

(b) What is the angle of reflection? **[1]**

Working

Answer °

(c) Ziad then shines a ray of light from the air into a tank of water. The ray changes direction at the surface of the water. What is this change of direction called? **[1]**

(d) Which of these is caused by refraction? **[1]**

- ☐ A) You see yourself in a bathroom mirror.
- ☐ B) A tree makes a shadow on the ground.
- ☐ C) A coin at the bottom of a glass of water looks higher than it really is.
- ☐ D) Light bounces off a shiny metal spoon.

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9 The museum has a rock collection. Ziad reads the information cards for three rocks. **[1]**

Rock	What it looks like
K	rounded grains stuck together in layers; some grains rub off; there is a fossil fish in it
L	large crystals of different colours that fit tightly together; no layers
M	very hard; small crystals lined up in wavy bands

(a) Which rock, K, L or M, is an **igneous** rock?

(b) Which rock, K, L or M, is a **sedimentary** rock? **[1]**

(c) Describe how the fossil fish in rock K could have formed. **[2]**

Paper 2 — 35 minutes — 40 marks

1 Nour's family goes camping in Sinai. **[1]**

(a) Nour stirs a spoonful of salt into a cup of hot water and another spoonful into a cup of cold water. The salt dissolves faster in the hot water. Which is the best explanation?

- ☐ A) The salt melts in the hot water.
- ☐ B) The water particles move faster in hot water, so the salt particles mix in among them more quickly.
- ☐ C) Hot water has fewer particles, so there is more space for the salt.
- ☐ D) The salt particles get bigger in hot water.

(b) Nour hangs wet towels on a rope in the sunshine. Two hours later the towels are dry. Name the process that made the water leave the towels. **[1]**

(c) Nour's brother says: "The water in the towels boiled away." Explain why he is wrong. **[2]**

2 The family cooks on a camping stove. Their cooking pot is made of steel. The handle of the pot is made of wood. **[1]**

(a) Why is the pot made of steel?

- ☐ A) Steel is a good thermal conductor, so heat passes quickly to the food.
- ☐ B) Steel is a thermal insulator, so it keeps the food cool.
- ☐ C) Steel is a poor conductor of electricity.
- ☐ D) Steel is a light material that floats on water.

(b) Explain why the handle is made of wood. **[1]**

(c) At school, Nour tests these objects in a circuit with a lamp to find out which ones are electrical conductors: **[1]**

a steel spoon, a wooden peg, a plastic cup, a strip of aluminium foil, a glass marble, a piece of copper wire.

How many of these objects will make the lamp light?

Working

Answer

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3 The high mountains of southern Sinai are made of granite. Granite is an igneous rock. **[1]**

(a) Granite formed deep underground when magma (hot liquid rock) cooled and turned into solid rock. What is the name of this process?

(b) Wind, rain and sudden floods break the granite into small grains of sand and carry the grains away. Name this process. **[1]**

(c) The sand grains settle in layers at the bottom of a lake. Over a very long time the layers are buried and pressed together into rock. Which type of rock forms? **[1]**

(d) Describe how this rock could change into a metamorphic rock. **[2]**

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4 Adam visits the doctor's clinic because he has a cough. The doctor explains how his lungs work. **[1]**

(a) Which list shows the path that air takes when Adam breathes in?

- ☐ A) nose → air sacs → trachea → bronchi
- ☐ B) trachea → nose → air sacs → bronchi
- ☐ C) nose → bronchi → trachea → air sacs
- ☐ D) nose → trachea → bronchi → air sacs

(b) Complete the sentence. **[1]**

In the air sacs of the lungs, oxygen moves from the air into the _____.

(c) The doctor says that many vertebrates have lungs like ours. Which animal takes in oxygen with lungs? **[1]**

- ☐ A) camel
- ☐ B) shark
- ☐ C) goldfish
- ☐ D) tuna

(d) Adam's cough is caused by a virus. The virus spreads in tiny drops when he coughs. Give **two** things Adam can do to stop the virus spreading to his family. **[2]**

5 In the school laboratory, Youssef investigates what makes iron nails rust. He sets up four test tubes and leaves them for 7 days. **[1]**

Tube	What is in the tube	Rust after 7 days
A	iron nail, air and tap water	some rust
B	iron nail in boiled water with a layer of oil on top, so no air can get in	no rust
C	iron nail in dry air (a drying powder removes the water from the air)	no rust
D	iron nail, air and salty water	a lot of rust

Before he started, Youssef predicted: "The nail in tube B will rust the most, because it is completely covered in water."

(a) Which **two** tubes show that water is needed for rusting?

- ☐ A) A and B
- ☐ B) B and C
- ☐ C) A and C
- ☐ D) C and D

(b) Rusting is a chemical reaction. Iron is one reactant. Name **one** other reactant. **[1]**

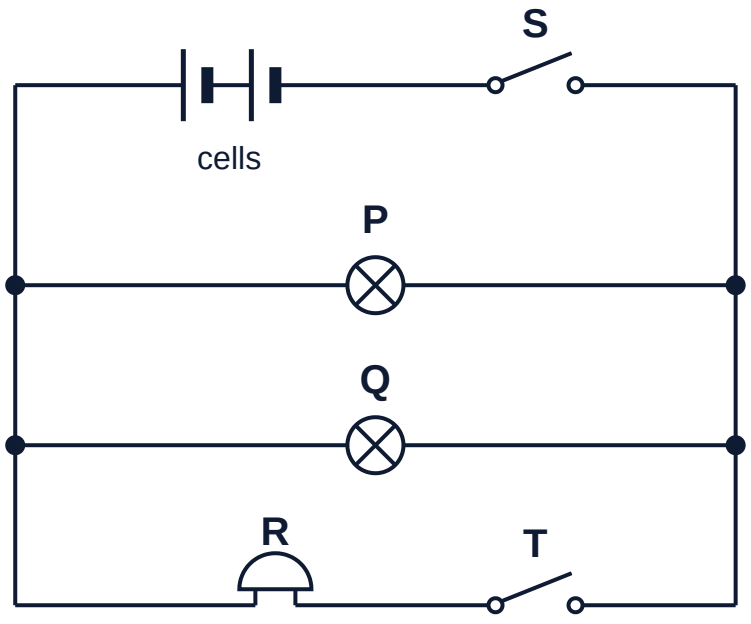
(c) What evidence shows that a chemical reaction happened in tube A? **[1]**

(d) Was Youssef's prediction correct? Use the results to explain your answer. **[1]**

(e) Give **one** variable Youssef should keep the same in all four tubes. **[1]**

(f) Youssef used only one nail in each tube. Suggest how he could make his results more reliable, and explain why. **[1]**

6 Rana designs a circuit for the lamps and the bell on her bicycle. The bell is a buzzer. Both switches are shown open. **[1]**



(a) Which letter labels the buzzer?

(b) Are lamps P and Q joined in series or in parallel? **[1]**

(c) Which switch or switches must be closed for the buzzer to sound? **[1]**

- ☐ A) S only
- ☐ B) T only
- ☐ C) S and T
- ☐ D) neither switch

(d) Switch S is closed. Lamp P breaks. What happens to lamp Q? **[1]**

- ☐ A) Lamp Q goes out.
- ☐ B) Lamp Q stays on.
- ☐ C) Lamp Q flashes on and off.
- ☐ D) Lamp Q gets much dimmer.

(e) Rana's old bicycle had the same cells, but its two identical lamps were joined in series. How does the brightness of the lamps in her new parallel circuit compare? **[1]**

- ☐ A) The lamps in the parallel circuit are brighter.
- ☐ B) The lamps are the same brightness in both circuits.
- ☐ C) The lamps in the parallel circuit are dimmer.
- ☐ D) The lamps in the parallel circuit do not light.

7 Rana's bicycle tyre is flat. She takes out the inner tube and puts it on a balance. Its mass is 205 g. She pumps air into the inner tube until it is hard and puts it on the balance again. Its mass is now 213 g. **[1]**

(a) What mass of air did Rana pump into the inner tube?

Working

Answer ____ g

(b) What does Rana's experiment show about gases? **[1]**

8 A display at the science museum gives the strength of gravity in four places. **[1]**

Place	Strength of gravity in N/kg
Earth	10
Moon	1.6
Mars	3.7
Jupiter	25

Ziad has a bag of sand with a mass of 2 kg.

(a) Work out the weight of the bag of sand on Earth.

Working

Answer ____ N

(b) What is the **mass** of the bag of sand on the Moon? **[1]**

Working

Answer ____ kg

(c) Work out the weight of the bag of sand on Mars. **[1]**

Working

Answer ____ N

9 At a pond, Mazen puts an empty plastic bottle with its lid on into the water. It floats. He adds sand to the bottle, 50 g at a time, and puts the lid back on each time. **[1]**

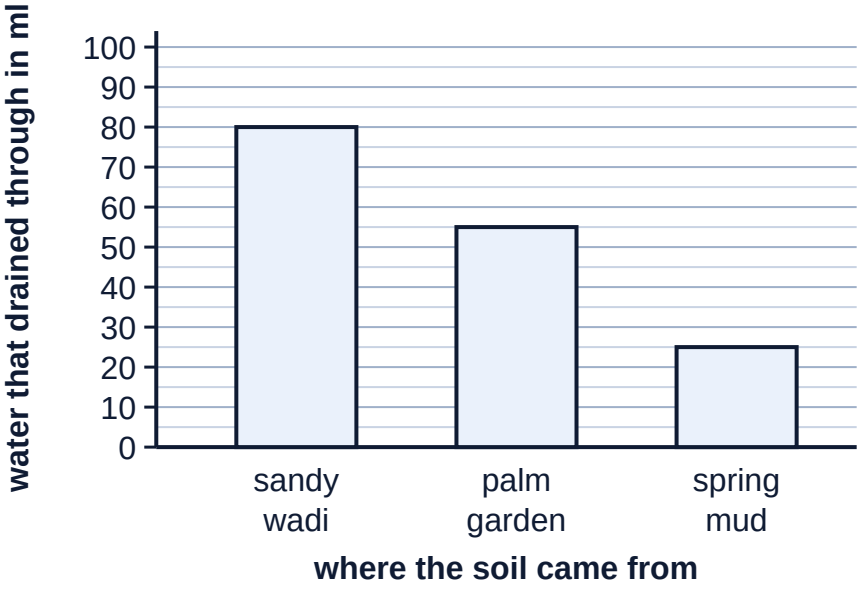
Mass of sand in g	Floats or sinks?
0	floats
50	floats
100	floats
150	floats
200	sinks

(a) Explain why the bottle sank when it had 200 g of sand in it.

(b) Mazen tips the 200 g of sand into a wide, empty plastic tub. The tub has the same mass as the bottle. The tub floats with the sand in it. Why? **[1]**

- ☐ A) The sand weighs less in the tub.
- ☐ B) The sand has dissolved in the water.
- ☐ C) The tub has less mass than the bottle.
- ☐ D) The tub has a different shape, so it pushes aside more water and gets more upthrust.

10 On the camping trip, Nour collects soil from three places: a sandy wadi (dry valley), a palm garden, and muddy ground by a spring. Back at school she puts 200 g of each soil in a funnel lined with filter paper. She pours 100 ml of water onto each soil and measures how much water drains through in 5 minutes. The bar chart shows her results. **[1]**



(a) How much water drained through the soil from the palm garden?

Working

Answer ml

(b) How much water did the spring mud **hold** (keep)? **[1]**

Working

Answer ml

(c) Which soil most likely has the **most clay** in it? **[1]**

- ☐ A) the spring mud
- ☐ B) the sandy wadi soil
- ☐ C) the palm garden soil
- ☐ D) they all have the same amount of clay

(d) A farmer wants to grow vegetables in the sandy wadi soil. Suggest one change she could make to the soil to help the plants grow, and explain why it helps. **[1]**

(e) Why did Nour draw a bar chart and not a line graph? **[1]**

- ☐ A) Bar charts are always more accurate than line graphs.
- ☐ B) She only had three results, and a line graph needs ten.
- ☐ C) The amount of water is not a number.
- ☐ D) The type of soil is a word (a category), not a number that is measured.